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AN UNPAIRED LEARNING-BASED METHOD FOR IMAGE DESPECKLING

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ABSTRACT

Speckle noise removal, or despeckling, is a crucial component of coherent imaging systems. In this work, we extend BD-QMAP [1], a theoretically grounded despeckling algorithm developed for stationary stochastic processes, to images and develop an image despeckling algorithm. Extensive experiments demonstrate that our approach outperforms classic Bayesian and MMSE despecklers. Furthermore, BD-QMAP, with its simple implementation, achieves performance close to that of classic image processing techniques and shows potential for further improvements to compete with supervised learning-based despeckling methods trained on large paired datasets—despite not requiring any paired ground truth and speckled data.

Index Terms— Bayesian despeckling, synthetic aperture radar, quantized MAP

1. INTRODUCTION

Coherent imaging systems, such as digital holography [2] and synthetic aperture radar [3], are known to suffer from *speckle noise* which degrades the quality of images they produce. The speckle noise is typically modeled as multiplicative noise: $Y_i = X_i W_i$, where Y_i is the noisy observation at time (or location) $i \in \mathbb{N}$, X_i is the ground-truth signal and W_i is independent and identically distributed speckle noise. In a fully developed speckle noise model, it is widely accepted to assume the speckle noise is Gaussian [4]. In this work, we propose an image despeckling algorithm based on a Bayesian despeckler with a solid theoretical foundation. It performs comparably to learning-based methods, while offering analytical tractability.

1.1. Related Work

Bayesian despeckling methods can be grouped into two main approaches: adaptive linear minimum mean squared error (MMSE) filtering and maximum a posteriori (MAP) estimation. Early work by Lee [5] introduced an affine approximation of the nonlinear speckle model, minimizing the ℓ_2 norm with matched first-order moments to develop an MMSE estimator. Later, Kuan [6] proposed a fully linear MMSE method, eliminating the approximation errors of the affine model. Both techniques rely on local statistics computed over fixed windows, leading to similar despeckling performance in practice [7].

Building on denoising methods for additive noise, several heuristic despeckling techniques have been widely used, especially for high-resolution imaging. These include non-local means [8], 3D block-matching [9], and neural network-based methods [10]. For example, the DnCNN architecture [11] has been trained for despeckling [12]. Self-supervised approaches, such as Noise2Noise [13] and Noise2Void [14], have been adapted for despeckling in [15, 16].

1.2. Notations

Size of a finite-element set $\mathcal{A}$, is denoted by $|\mathcal{A}|$. $[x]_b = 2^{-b}\lfloor 2^b x \rfloor$ represents the b-bit quantized version of $x \in \mathbb{R}$ for any $b \in \mathbb{N}^+$. For $x^k \in \mathcal{X}^k$, $[x^k]_b$ denotes the element-wise b-bit quantization of x^k. The k-th order empirical distribution of $u^n \in \mathcal{U}^n$, $\hat{p}^k(\cdot|u^n)$, is defined as follows. For $a^k \in \mathcal{U}^k$,

$$\hat{p}^k(a^k|u^n) = \frac{1}{n-k+1} \sum_{i=1}^{n-k+1} \mathbb{1}_{u_i^{i+k-1}=a^k}. \tag{1}$$

2. BAYESIAN IMAGE DESPECKLING VIA BD-QMAP

BD-QMAP [1] is a theoretically grounded Bayesian despeckling method for stationary stochastic processes. It is inspired by QMAP denoising originally proposed for additive white Gaussian noise removal [17]. Intuitively, given noisy sequence Y^n, BD-QMAP aims to reconstruct a sequence that is consistent with both the measurements and the source structure.

Consider a structured stationary random process $\mathbf{X} = \{X_i\}_{i\in\mathbb{N}}$, i.e., a stationary process with information dimension strictly less than one [18]. Let $\mathcal{X} \subset \mathbb{R}$ denote the alphabet of source $\mathbf{X}$. Given quantization level $b \in \mathbb{N}^+$, let $\mathcal{X}_b = \{[x]_b : \ x \in \mathcal{X}\}$ denote the b-bit quantized version of

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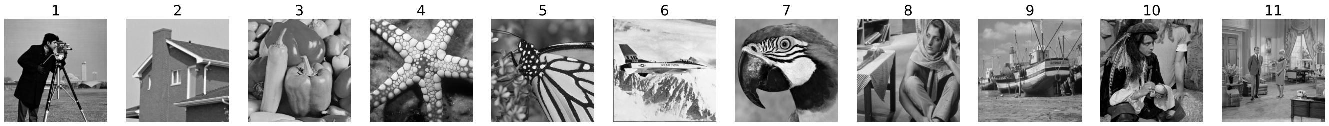

Fig. 1. Set 11 test images used for despeckling. Resolution for images 1 to 7 is 256×256 and 512×512 for the rest.

Table 1. Image Despeckling Performance Comparison in $\mathrm{PSNR} = -10\log(\mathrm{MSE}^2)$ for Grayscale Images Shown in Figure 1.

	1	2	3	4	5	6	7	8	9	10	11	average
Speckled	9.54	8.80	9.51	9.25	10.15	6.75	9.88	9.82	9.29	10.39	9.89	9.39
Box Car	16.88	17.31	17.11	16.86	17.14	14.85	16.95	17.48	17.35	18.48	17.78	17.11
Kuan [6] / Enhanced [7]	18.93/19.91	19.36/20.68	19.27/20.20	19.01/19.89	19.29/19.93	17.31/18.20	19.21/19.99	19.33/20.23	19.32/20.53	20.52/21.70	19.77/20.92	19.21/20.20
SAR-BM3D [9]	22.61	24.82	22.84	21.27	22.19	20.91	21.27	23.12	23.56	24.26	23.45	22.75
ID-CNN [12]	23.90	25.26	23.58	21.43	23.48	21.69	23.94	22.49	24.24	24.88	24.18	23.55
BD-QMAP$_b$ (4×4)	20.82	22.56	21.05	20.38	21.12	19.82	20.35	20.75	21.11	22.38	21.47	21.07
BD-QMAP$_b$ (7×7)	21.02	22.13	21.52	20.47	20.66	19.36	20.16	21.53	22.88	23.65	22.73	21.46

$\mathcal{X}$. Given $a^k \in \mathcal{X}_b^k$, a b-bit quantized k-tuple, BD-QMAP defines $w_{a^k} > 0$, the weight associated to a^k as

$$w_{a^k} = -\log \mathrm{P}([X^k]_b = a^k). \tag{2}$$

Here, the probability is calculated with respect to the distribution of process $\mathbf{X}$. Using the collection of weights $\mathbf{w} = (w_{a^k} : a^k \in \mathcal{X}_b^k)$, we define the weight associated to $u^n \in \mathcal{X}_b^n$ as

$$c_{\mathbf{w}}(u^n) = \sum_{a^k \in \mathcal{X}_b^k} w_{a^k} \hat{p}^k(a^k|u^n). \tag{3}$$

The function $c_{\mathbf{w}}(u^n)$ captures the structural consistency of candidate reconstruction sequence u^n with the source $\mathbf{X}$. Using this measure along with the likelihood function as a fidelity constraint to the speckled observation, BD-QMAP estimates X^n as $\hat{X}^{n,(k,b)}$ as follows

$$\hat{X}^{n,(k,b)} = \operatorname*{arg\,min}_{u^n \in \mathcal{X}^n} \frac{1}{n} \sum_{i=1}^{n} \big(\log u_i^2 + \frac{Y_i^2}{u_i^2}\big) + \frac{\lambda}{b} c_{\mathbf{w}}(u^n). \tag{4}$$

Here b and k denote the quantization level and the memory parameter (or the patch size), respectively.

2.1. Patchwise BD-QMAP for Images

Solving (4) directly is computationally prohibitive, as it involves a loss function that combines continuous and discrete components optimized over a continuous space. To make the problem more tractable, we restrict the search space to b-bit quantized sequences. For 1-dimensional sequences, this restricted sub-optimal optimization, which we refer to as BD-QMAP$_b$, can be solved using the Viterbi algorithm [19]. For piecewise constant sources, [1] demonstrates after solving BD-QMAP$_b$, the solution can be further refined using its structure and solving (4) for the weights (with no quantization). On the other hand, the source distribution for images is not available, and needs to be estimated (or learned) from training samples as

$$\hat{w}_{a^{k^2}} = -\log \hat{\mathrm{P}}([X^{k^2}]_b = a^{k^2}), \tag{5}$$

where $\hat{\mathrm{P}}([X^{k^2}]_b = a^{k^2})$ denotes the number of occurrences of quantized patch a^{k^2} divided by the number of training patches.

For image despeckling, even after restricting the search space to the set of quantized sequences, still solving BD-QMAP$_b$ is computationally challenging. The number of states of the Viterbi algorithm is proportional with the number of possible quantized sequences of length k^2, i.e., $|\mathcal{X}_b|^{k^2}$, which becomes infeasibly large, even for small image patches. Therefore, instead of solving (3) for the whole image, we first despeckle each $k \times k$ patch in the image individually, and then reconstruct the whole image by averaging the despeckled patches according to their locations in the image. This patch-wise optimization of (4) while suboptimal, as explored in Section 3, still achieves competitive despeckling performance.

Another issue that needs to be addressed is the problem of learning the weights from training patches. Again, the key challenge is the number of weights, which grows exponentially with k^2 and b. However, as shown in [17], for structured sources, the weights can be divided into two groups. A smaller set of key weights that determine the source structure and a considerably larger set of weights that are negligible, i.e., correspond to patches that have very low probability of occurrence. Using this key observation, following [17], we use a small fully connected neural network with binarized latent code to efficiently learn those key weights. The details are described in Section 3.1.

3. EXPERIMENTS

We conduct our experiments on the widely known test set of eleven images shown in Figure 1. As a measure of fidelity of the estimated despeckled image to its ground-truth we use PSNR and SSIM [20] which are reported in the Tables 1, 2, respectively. For visual comparison Figure 2 shows the reconstruction quality of each despeckler for an image of the test set. The details on the implementation of each algorithm

Table 2. Image Despeckling Performance Comparison in SSIM [20] for Grayscale Images Shown in Figure 1.

	1	2	3	4	5	6	7	8	9	10	11	average
Speckled	0.20	0.07	0.13	0.14	0.19	0.09	0.22	0.13	0.11	0.10	0.12	0.14
Box Car	0.39	0.37	0.49	0.51	0.50	0.28	0.49	0.41	0.40	0.42	0.39	0.42
Kuan [6] / Enhanced [7]	0.35/0.40	0.25/0.32	0.39/0.46	0.43/0.50	0.45/0.51	0.22/0.27	0.45/0.50	0.34/0.41	0.30/0.37	0.32/0.40	0.31/0.38	0.35/0.41
SAR-BM3D [9]	0.61	0.62	0.63	0.58	0.67	0.51	0.63	0.61	0.56	0.59	0.56	0.60
ID-CNN [12]	0.64	0.59	0.68	0.58	0.71	0.50	0.70	0.55	0.57	0.55	0.56	0.60
BD-QMAP$_b$ (4×4)	0.46	0.44	0.54	0.55	0.60	0.51	0.55	0.45	0.42	0.45	0.44	0.49
BD-QMAP$_b$ (7×7)	0.57	0.47	0.60	0.56	0.60	0.47	0.63	0.51	0.54	0.52	0.51	0.54

in the two tables are provided in the rest of this Section. It is worth noting that BD-QMAP algorithm only requires a set of clean patches of size $k \times k$ to learn the weights in (4), in contrast to supervised neural network despecklers like ID-CNN [12] which require a large dataset including the pairs of clean and speckled images.

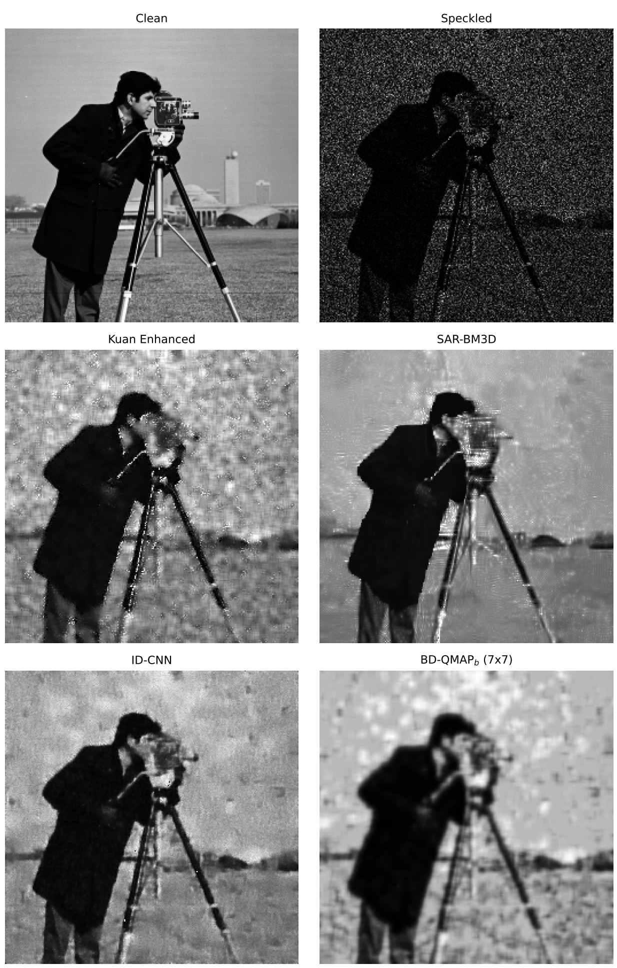

Fig. 2. Despeckling reconstruction for visual comparison of *cameraman* image (1).

3.1. BD-QMAP Setup

To learn the set of weights using (5) for vectorized patch inputs of size k^2, we employ a fully connected autoencoder as shown in Figure 3, with a single layer of size $n_b = 32$ as encoder and three layers of size $n_1, n_2 = 200$ and $n_3 = k^2$ as decoder. The activation function for the latent code is Sign, while other layers use ReLU as the nonlinearity. To enable backpropagation with non-zero gradients for updating the encoder's parameters, we use the gradients of $\tanh$ [21] and minimize the mean squared error as the loss function. After training the autoencoder, the weights in (5) are learned by estimating the empirical distribution of the binary representations of dataset patches.

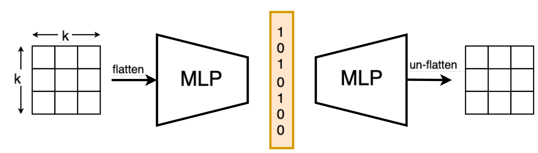

Fig. 3. Finding binary representations of patches for learning weights in Equation (5). MLP denotes a multi-layer perceptron neural network.

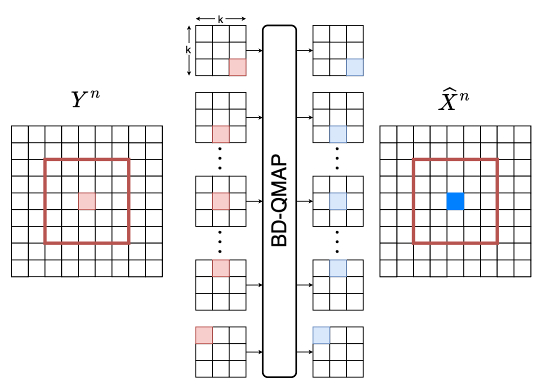

Fig. 4. Patch-wise despeckling of a single pixel using all patches of size $k \times k$ including the pixel in hand.

The BD-QMAP method requires selecting a hyperparameter λ. Based on numerical experiments, we set $\lambda = 0.1$ for a patch size of $k = 4$ and $\lambda = 0.01$ for a patch size of $k = 7$.

After determining the weights in Equation (5), we proceed to solve the BD-QMAP optimization problem in (4) for all

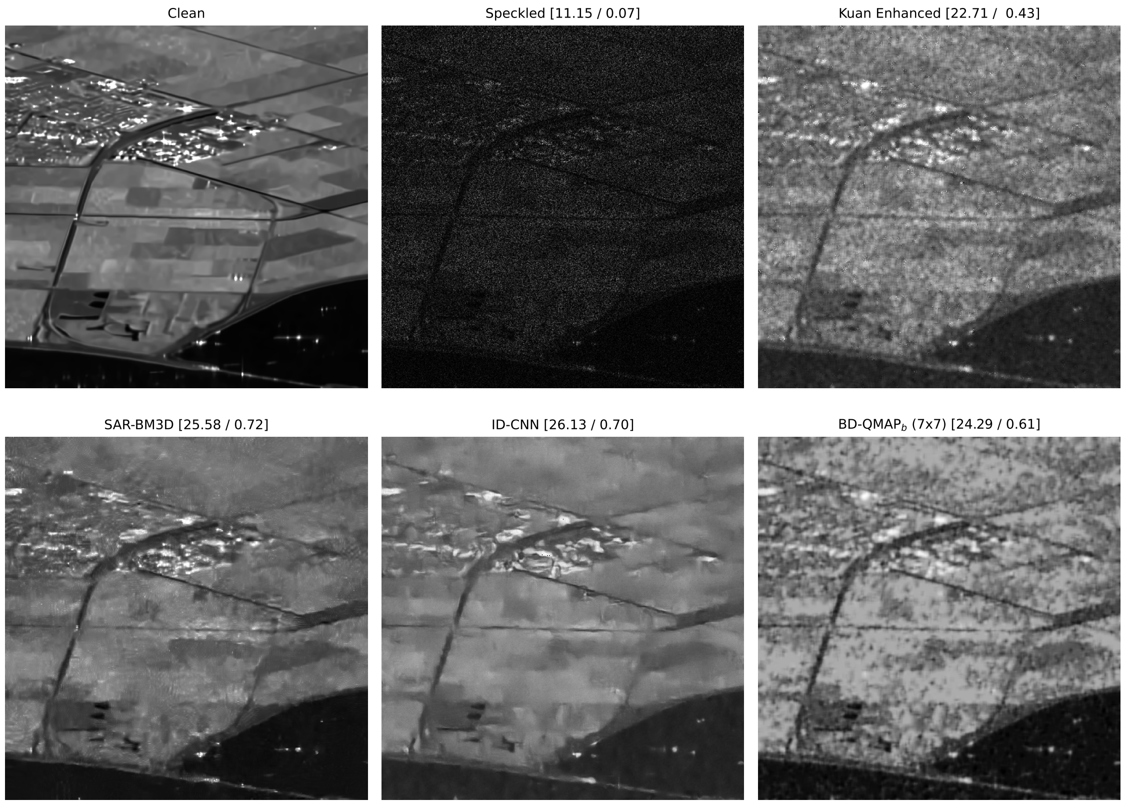

Fig. 5. Despeckling reconstruction for visual comparison of *Lely* image (500x500) [22]. PSNR/SSIM metrics are reported above each image.

extracted $k \times k$ patches in the image. Consequently, each despeckled pixel is obtained by averaging all the despeckled estimates of that pixel from the initial run, as illustrated in Figure 4.

3.2. Linear adaptive filters and learning-based Methods

All linear adaptive filters in Tables 1 and 2 need a window-size hyperparameter to determine the region for computing local statistics. We empirically find setting local window to seven being optimal in terms of the quality measures PSNR and SSIM [20]. We also test *enhanced* versions of these linear adaptive filters by adjusting them for different types of regions, following the method in [7]. This method categorizes areas as homogeneous, heterogeneous, or strongly speckled based on the local coefficient of variation, given by $C_Y = \mathrm{Var}(Y)/\mathbb{E}[Y]$. The enhanced filter works as follows: If C_Y is smaller than the noise variation coefficient $C_W = \mathrm{Var}(W)/\mathbb{E}[W]$, the filter replaces the pixel with the average value of the surrounding area. If C_Y is greater than or equal to $C_{\max} = \sqrt{3}C_W$, the filter does nothing, keeping the original pixel. Otherwise, the filter is applied as usual.

As a supervised learning-based despeckler, we trained ID-CNN [12], which follows the architecture of the well-known additive denoiser DnCNN [11]. The ID-CNN was trained from scratch using a synthesized dataset of 400 images selected from BSD [23], as described in [24]. Each dataset pair consists of a ground-truth image and its speckle-corrupted version, where the speckle noise follows a standard Gaussian distribution. The model was trained for 100 epochs with a batch size of 64, using the Adam optimizer with a learning rate of 0.001.

4. CONCLUSION

In this work, we proposed a new image despeckling method by extending BD-QMAP, a despeckling algorithm, originally designed for structured stationary processes. We demonstrated that a suboptimal extension of the algorithm to high-dimensional images outperforms well-established MMSE and Bayesian filters while achieving performance comparable to learning-based methods—without requiring paired datasets of clean and noisy images. Furthermore, our approach is inherently adaptable to multi-look despeckling without the need for retraining, making it a flexible and data-efficient solution for a wide range of despeckling applications.

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